

D.4 Induction [HL only]

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **25 marks**

Show your working.

Question 1 — The slow-motion magnet (8 marks)

A strong magnet dropped down a vertical *copper* pipe takes several seconds to emerge; an identical-looking steel bolt falls straight through.

- (a) Define magnetic flux through a surface. [2]
- (b) Use Faraday's and Lenz's laws to explain the magnet's leisurely descent. [4]
- (c) State the energy source of the currents induced in the pipe, and what the electrical energy finally becomes. [2]

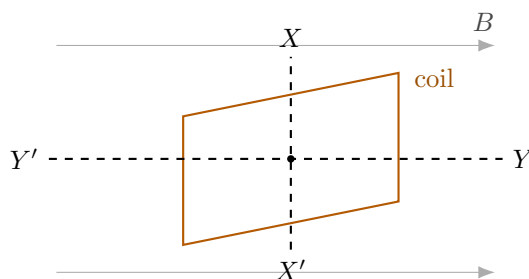
Question 2 — The switch-off spike (7 marks)

A flat coil of 200 turns and area $4.0 \times 10^{-3} \text{ m}^2$ sits inside a field of 0.50 T directed through its plane's normal. The field is switched off, falling steadily to zero in 0.10 s.

- (a) Calculate the initial flux linkage of the coil. [2]
- (b) Calculate the average emf induced during the switch-off. [3]
- (c) Before the switch-off, while the 0.50 T field was steady, the induced emf was zero. State the principle this illustrates, in one sentence. [2]

Question 3 — The generator that generated nothing (10 marks)

A rectangular coil can be spun inside a uniform magnetic field, about either of two axes: axis XX' lies in the plane of the coil and perpendicular to the field; axis YY' is parallel to the field.



- (a) The coil is spun steadily about XX' . Describe how the flux through the coil varies, and sketch the shape of the emf against time. [3]
- (b) State the positions of the coil (relative to the field) at which the emf is (i) maximum and (ii) zero, and explain the apparent paradox that the emf peaks when the flux is zero. [4]
- (c) An apprentice instead spins the coil about YY' and measures nothing at all. Explain why the emf is exactly zero for this axis. [3]

Answers

Question 1 — The slow-motion magnet

- (a) $\Phi = BA \cos \theta$: the field strength times the area times the cosine of the angle between the field and the *normal* to the surface. Equivalently, the component of B perpendicular to the surface, multiplied by the area.
- (b) As the magnet falls, the flux through each ring of copper pipe changes, and by Faraday's law that changing flux induces an emf and hence eddy currents in the copper. By Lenz's law those currents flow so as to *oppose the change* that produced them: the pipe ahead of the magnet repels it and the pipe behind attracts it, giving a persistent retarding force. The steel bolt is not a magnet, so it produces no flux change, no eddy currents and no braking, and it falls freely. (Copper's high conductivity is what makes the eddy currents, and so the braking, large.)
- (c) The gravitational potential energy of the falling magnet. It finishes as internal energy, heating the copper, dissipated against the pipe's resistance.

Question 2 — The switch-off spike

- (a) $N\Phi = NBA = 200 \times 0.50 \times 4.0 \times 10^{-3} = 0.40 \text{ Wb turns}$.
- (b) $\varepsilon = \frac{\Delta(N\Phi)}{\Delta t} = \frac{0.40}{0.10} = 4.0 \text{ V}$.
- (c) Induction requires a *changing* flux, not a large one. A steady field, however strong, gives $d\Phi/dt = 0$ and induces nothing.

Question 3 — The generator that generated nothing

- (a) Spinning about XX' swings the coil's normal round relative to the field, so the flux varies sinusoidally between $+BA$ and $-BA$. Since $\varepsilon = -N d\Phi/dt$, the emf is also sinusoidal, but a quarter of a cycle out of phase with the flux: it peaks where the flux graph is steepest and falls to zero where the flux graph turns over.
- (b) (i) The emf is *maximum* when the coil's plane is *parallel* to the field, equivalently its normal perpendicular to B : the flux is momentarily zero but sweeping through zero at its fastest rate.
(ii) The emf is *zero* when the coil's plane is *perpendicular* to the field, its normal along B : the flux is at its maximum BA but, being at a turning point, momentarily unchanging. There is no paradox, because the emf tracks the *rate of change* of flux, not the amount of it. A large quantity sitting at a maximum is changing slowly; a quantity passing through zero can be changing very fast.
- (c) Rotating about YY' leaves the angle between the coil's normal and the field unchanged, because the axis is itself parallel to the field. So $\Phi = BA \cos \theta$ keeps the same value at every instant, $d\Phi/dt = 0$, and the emf is exactly zero no matter how fast the coil is spun. Rotation alone does not induce an emf; only a *change of flux* does.